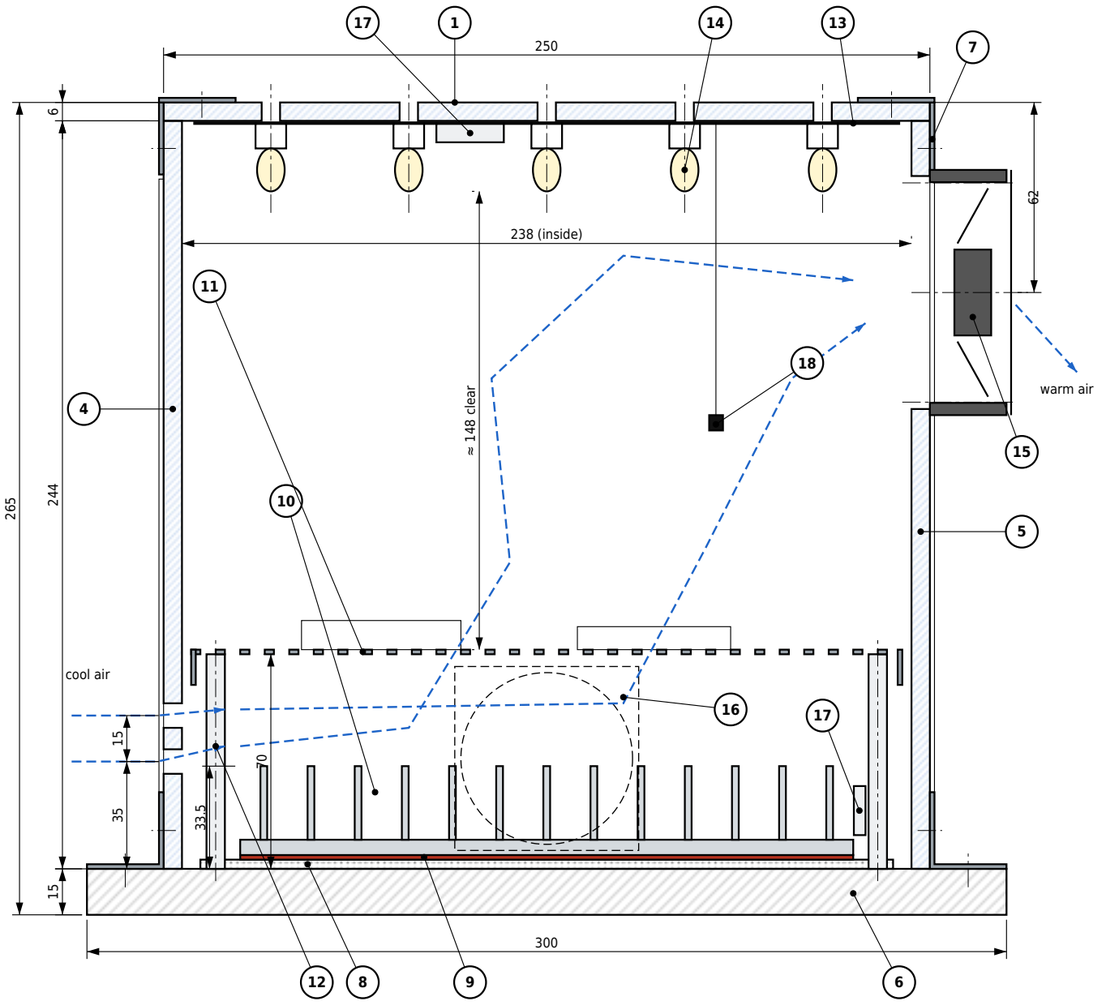
SECTION A-A · SCALE 1:2

Cut at mid-depth
(front removed;
looking at the back)



ASSEMBLY NOTES

1. 6 mm clear solid polycarbonate: continuous use up to ~115 °C. TH1 (85 °C) and the firmware cut-off (78 °C) leave margin.
2. Cut with a fine-tooth blade (≥ 60 teeth) or an acrylic blade, with the protective film on. Sand and break the edges 0.5 mm.
3. Drill with a sharp bit at low speed, wood backing underneath. A holes are Ø5 for M4: clearance for PC expansion (~1 mm over 250 mm from 20 to 80 °C).
4. Do not use solvent cement, cyanoacrylate or cutting oil: they stress-crack PC. Seal gaps with neutral high-temperature RTV silicone.
5. Angles outside on the 8 fixed edges; the front stays free for the door. Stainless M4 with a washer on both sides, light torque.
6. Order: bottom angles on the base → back and sides → back verticals → top and top angles → door, catch and seal.
7. Heater bonded under the heatsinks with thermal paste, on the insulation. TH2 screwed to the heatsink end. No PC within 15 mm of the fins.
8. M2 on the back, under the shelf, blowing along the fins. The shelf comes out through the door, resting on the 4 spacers.
9. TH1 under the top; LM35 hanging ~75 mm above the shelf, out of the lamp beam. Wires exit through the cable glands on the back.
10. Airflow: in through the left-side vents (low), across the heatsinks, up through the shelf and out through fan M1 (right side, high).
11. Remove the protective film only at the end of assembly.

REVISIONS

REV.	DESCRIPTION	DATE	BY
A	Initial release: 6 mm PC chamber 250 × 250 × 250 (Rev. D of PIDL-ESQ-001).	11/10/2026	GM

BILL OF MATERIALS (MECHANICAL)

ITEM	DESCRIPTION	RATING	QTY
1	Top · clear solid PC	250 × 250 × 6	1
2	Back · PC	250 × 244 × 6	1
3	Door · PC	250 × 243 × 6	1
4	Left side (vents) · PC	238 × 244 × 6	1
5	Right side (fan) · PC	238 × 244 × 6	1
6	MDF base + 4 rubber feet	300 × 300 × 15	1
7	Aluminium angle (see sheet 2)	25 × 25 × 1.5	8
8	Ceramic-fibre insulation	226 × 226 × 3	1
9	Silicone heater H1 (PIDL-ESQ-001)	200 × 200 · 80 W	1
10	Finned aluminium heatsink	200 × 100 × 30	2
11	Perforated aluminium shelf	232 × 232 × 1.5	1
12	Spacer: Al tube + M4 × 90 screw + tee nut	Ø8 × 70	4
13	Aluminium reflector	230 × 230 × 0.5	1
14	T10 socket + W5W lamp (EL1-EL5)	12 V · 5 W	5
15	Exhaust fan M1 + grille	80 × 80 × 25	1
16	Circulation fan M2	60 × 60 × 15	1
17	KSD301 thermostat (TH1 air, TH2 heatsink)	85 °C · 120 °C	2
18	LM35DZ sensor (TS1)	TO-92	1
19	Stainless corner hinge	40 mm	2
20	Handle	64 mm centres	1
21	Magnetic cabinet catch	—	1
22	Silicone D-profile seal	9 × 8 · 1 m	1
23	PG7 cable gland	cable Ø3-6.5	2
24	Stainless M4 × 12 screw + lock nut + 2 washers	A2	64
25	Screws M4 × 35 (M1) · M3 × 30 (M2) · M3 × 10 (TH1)	A2	4+4+2
26	Wood screw (angles to base)	3.5 × 16	12
27	High-temp RTV silicone + thermal paste	≥ 250 °C	1

THERMAL SIZING

Inside volume	238 × 238 × 244 mm = 13.8 L
Shelf area	232 × 232 mm, ~148 mm clear
Heat loss	UA ≈ 1.6 W/K (PC U ≈ 3.9 W/m²K + base)
Capacity · time constant	≈ 1640 J/K · τ ≈ 1000 s
Power	5 × 5 W lamps + 80 W heater = 105 W
Steady ΔT (100 %)	lamps only +16 °C · with heater +66 °C
Simulated (25 °C amb.)	25→45 °C 12 min · 45→70 °C 20 min · 70→35 °C 8 min
Heatsink	max. ≈ 89 °C at 70 °C air (TH2 opens at 120 °C)

	NAME	INIT.	DATE
DRAWN	Gabriel Marques	GM	11/10/2026
APPROVED	Gabriel Marques	GM	11/10/2026

PROJECT PIDlampada

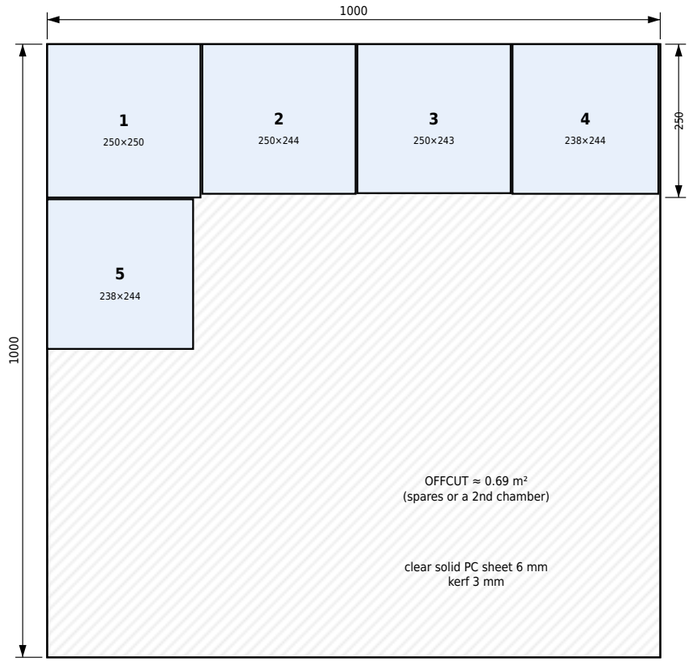
THERMAL CHAMBER WITH PWM CONTROL

ASSEMBLY · SECTION A-A · BILL OF MATERIALS

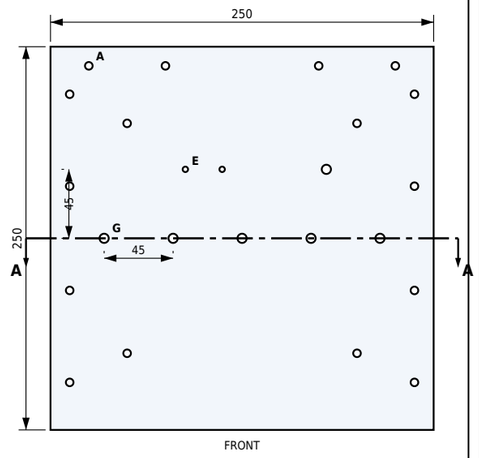
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DRAWING No.	REV.	SHEET	SCALE	SIZE	DATE
PIDL-MEC-001	A	1 / 2	1:2	A3	11/10/2026

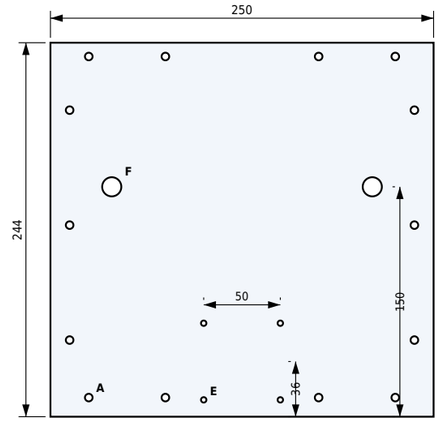
CUTTING PLAN · 1:10



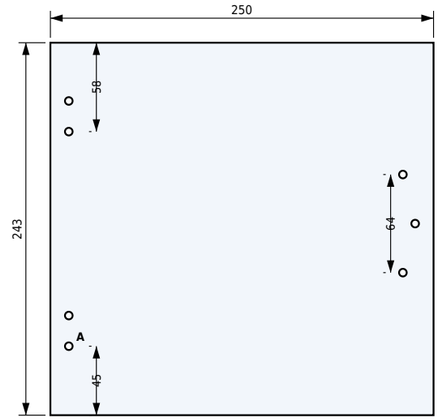
[1] TOP · 1:4



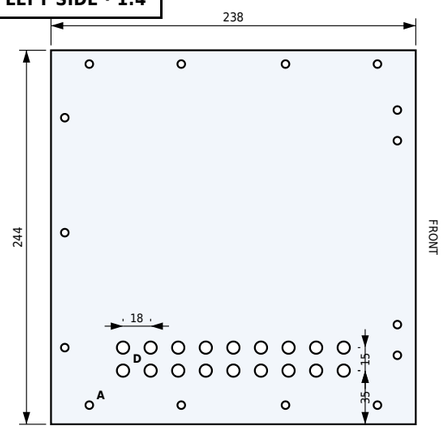
[2] BACK · 1:4 (seen from behind)



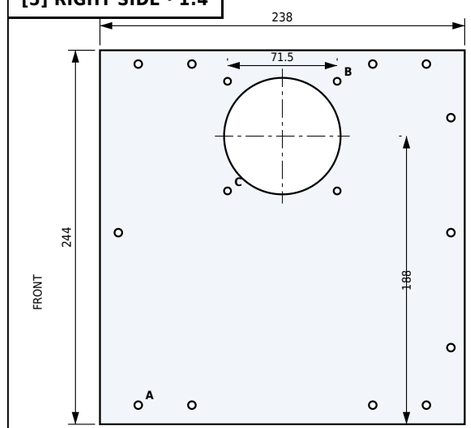
[3] DOOR · 1:4



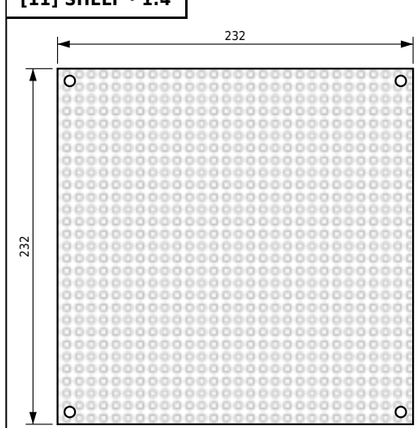
[4] LEFT SIDE · 1:4



[5] RIGHT SIDE · 1:4

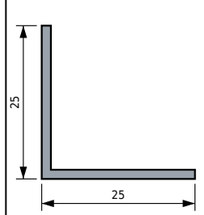


[11] SHELF · 1:4

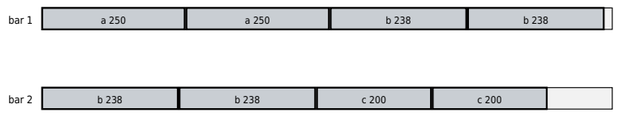


Al 1.5 mm · Ø5 holes, 8 pitch (40 %)
4 × Ø4.5 at 8 mm from corners · 10 mm flange

[7] ANGLES



PROFILE 1:1
thk. 1.5



2 bars of 1 m · scale 1:10

POS.	LENGTH	QTY	WHERE
a	250	2	back, bottom and top
b	238	4	sides, bottom and top
c	200	2	back verticals

HOLE TABLE

HOLE	Ø	QTY	PANELS	PURPOSE
A	5.0	64	1 2 3 4 5	M4: angles, hinges, handle, catch, reflector
B	76	1	5	fan M1 opening
C	4.5	4	5	M1 mounting (71.5 square)
D	8.0	18	4	intake vents (2 × 9, 18 pitch)
E	3.5	6	1 2	M3: M2 (50 square) and TH1 (24 centres)
F	12.5	2	2	PG7 cable glands
G	6.0	6	1	lamp (5) and LM35 (1) wires

Dimensions in mm. Holes without a position dimension: see the coordinates in desenho_mecanico.py (FUROS dict).

	NAME	INIT.	DATE
DRAWN	Gabriel Marques	GM	11/10/2026
APPROVED	Gabriel Marques	GM	11/10/2026

PROJECT PIDlampada

THERMAL CHAMBER WITH PWM CONTROL

CUTTING PLAN · PANEL DRILLING

UNIP · COMPUTER ENGINEERING · CAPSTONE PROJECT 2026 — SÃO JOSÉ DOS CAMPOS - SP · BRAZIL

DRAWING No.	REV.	SHEET	SCALE	SIZE	DATE
PIDL-MEC-001	A	2 / 2	AS SHOWN	A3	11/10/2026